



Aprendizagem Automática 2023/2024

A bird's eye view of supervised learning

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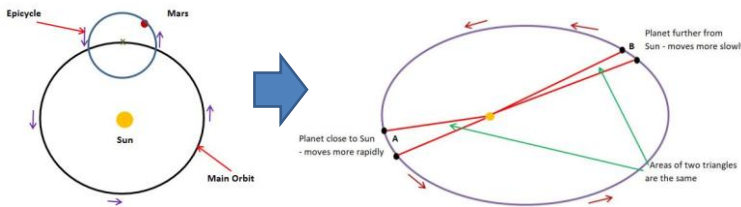
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Summary

- What is machine learning (again)
- Main objectives
- Main challenges
- A classification problem from top to bottom
 - The nature of classification
 - Selecting a model and fitting it
 - Evaluating and validating it
- A regression problem
 - The nature of regression
 - Selecting a model and fitting it
 - Evaluating and validating it
- Conclusions and further notes

Copernicus, Kepler and Newton (and Tycho Brahe!)

- The Heliocentric model of the planets
- The Copernican approach and the Epicycles
- The Elliptical model of Kepler
- And Newton wraps it up (well sort of...)



Creating the final model

$$T^2 = \frac{4\pi^2}{GM} a^3$$

can be expressed
as simply

$$T^2 = a^3$$

If expressed in the following units:

T Earth years

a Astronomical units AU
($a = 1$ AU for Earth)

M Solar masses $M_{\odot}$

$$\text{Then } \frac{4\pi^2}{G} = 1$$

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3$$

<http://hyperphysics.phy-astr.gsu.edu/hbase/kepler.html>

Main Objectives

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- [Of this course? of Machine Learning?, of life itself?]
- To go from the **known and easy** to the **unknown and hard**
- Also to be able to understand the world better
 - Is the model understandable? What does it tell us about the world
- Two different approaches
 - When we know what we are looking for:
 - **Supervised Modeling**
 - When we do not:
 - **Unsupervised Learning**



Main objective of this course!

What is supervised learning?

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- The idea is learning from examples
- Uses a set of data for which we know everything
 - A set of measured **independent** variables (**the Xs**)
 - One **dependent** variable (**the y**) – This one typically harder or more expensive to obtain
- determine the best **f()**, for computing **y'=f(X)**
 - such that the difference between **y** and **y'** is as small as possible. E.g.
 - predicting if an email is SPAM or not
 - We could use as a metric the ratio of many right predictions we made
 - predicting the weight of anyone given their age and height
 - We could minimize the mean sum of the errors

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Supervised Learning

- Examples

- Predict whether or not one person has Cancer, given a lot of biometric and serological data from a set of people from which the diagnosis exists

- y – Boolean (Cancer or no-cancer)
- X – biometric data and serological data for each person

Easy problem to model

- Predict the Economy growth as a fraction of previous GDP using economic, political and structural data from each country

- y – GDP growth (real number)
- X – Economic data (both quantitative and qualitative) for each country

Not so easy problem to model

- Predict the likelihood someone will buy something in a web store given their click behaviour and responses to advertisement

- y – "likelihood of buying" (?)
- X – "Click behaviour" (?) and "responses to advertisement" (?)

Harder problem to model

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Supervised learning

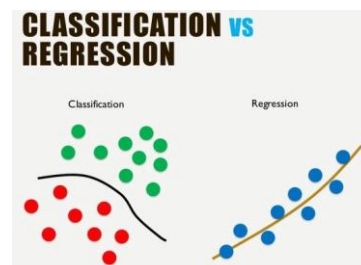
- The problem with supervised modeling is that the **relationship between dependent and independent variables is typically non linear**

- The "**best**" possible model is an elusive concept

- Depends on the data
- Depends on the representation of the data

- Ultimately the problem is at least **NP-Hard**

- Heuristics have to be used
- All machine learning algorithms are just heuristics that assume some sort of behavior on the part of data or make compromises on the models produced



<https://chatbotsmagazine.com/lets-know-supervised-and-unsupervised-in-an-easy-way-9168363e06ab>

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Main Challenges

- The hardness of the supervised learning process
- The Curse of Dimensionality
- The No-Free-Lunch Theorem
- Balancing bias and variance
 - The problem of over-fitting
 - How much can we abuse a model

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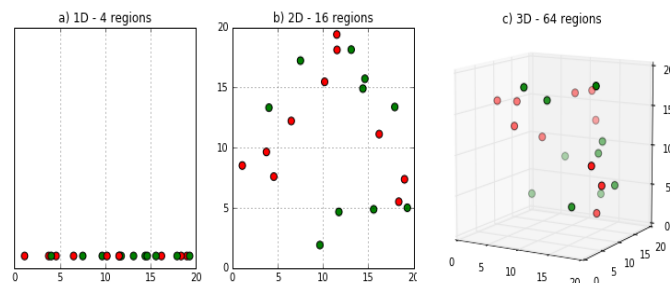
Complexity of Supervised Learning

- Unless very specific assumptions are made (e.g. [Linearity of the relations](#)) - Supervised Learning is a **NP-Hard problem**
- What does it mean?
 - Basically to find the optimal solution has an exponential time complexity relative to the size of data and verifying it requires analyzing each possible solution
 - Essentially it means that **even if we have a “very good model” we may never know if it is the best or how close we are to the best**
- **There is no solution to this problem**
 - There are only simplifying assumptions that may help us in finding good models. But even with those simplifications, we are most of the times not guaranteed to find the best solution

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The curse of dimensionality

- The curse of dimensionality (Bellman, 1961)
 - The larger the number of dimensions, the more likely it is that each sample is closer to the edge of the N-Dimensionality space than to any other sample



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Anomalies in high dimensional data

- Concentration of scores and distances: derived values such as distances become numerically similar
- Irrelevant attributes: in high dimensional data, a significant number of attributes may be irrelevant
- Definition of reference sets: for local methods, reference sets are often nearest-neighbor based
- Incomparable scores for different dimensionalities: different subspaces produce incomparable scores
- Interpretability of scores: the scores often no longer convey a semantic meaning
- Exponential search space: the search space can no longer be systematically scanned
- Data snooping bias: given the large search space, for every desired significance a hypothesis can be found
- Certain objects occur more frequently in neighbor lists than others.

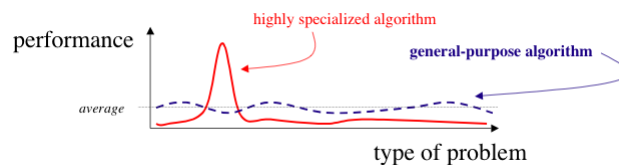
Zimek et al 2012 - <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11161>

The No-Free Lunch Theorem

No-free-lunch theorem

(Wolpert and McReady, 1987) :

"If an algorithm performs better than random search on some class of problems then it must perform worse than random search on the remaining problems"



<https://towardsdatascience.com/a-blog-about-lunch-and-data-science-how-there-is-no-such-a-thing-as-free-lunch-e46fd57c7f27>

Balancing Bias and Variance

- Bias – difference between the average prediction and the correct value. High bias, means that the model does not focus on each sample rather on global trends. These models may have somewhat high errors but tend to generalize well
- Variance – is the variability of model predictions for a given point. Models with high variance focus too much on each individual sample. These lead to models that fit extremely well but generalize poorly
- Expected Error

$$Err(x) = E[(Y - \hat{f}(x))^2]$$

$$Y = f(x) + \sigma_c^2$$

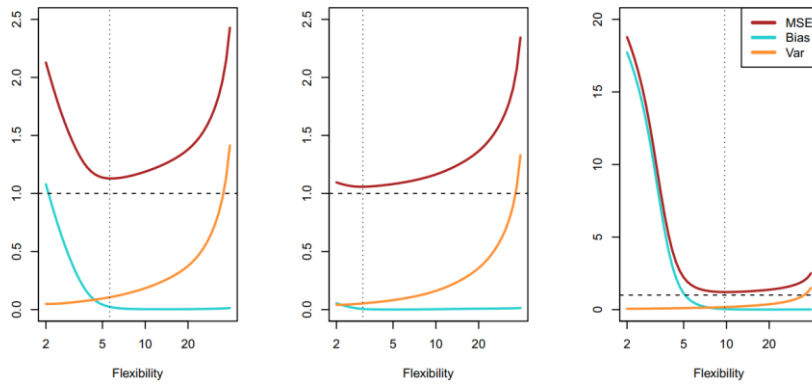
$$Err(x) = (E[\hat{f}(x)] - f(x))^2 + E\left[(\hat{f}(x) - E[\hat{f}(x)])^2\right] + \sigma_c^2$$

$$Err(x) = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

<https://towardsdatascience.com/understanding-the-bias-variance-tradeoff-165e6942b229>

Bias and Variance

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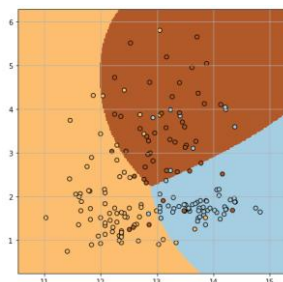
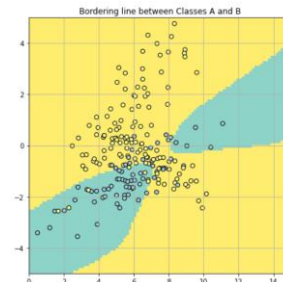


From James et al ISLP, 2023

Classification

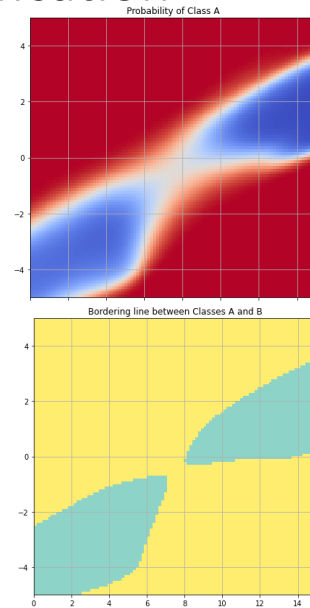
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- The main idea is to find a **separation between two or more different populations**
- Binary classification (separation between just 2 classes) is a sub-problem
 - Yet, most algorithms perform multiclass classification as several binary classification problems!
- Regression methods are not adequate even for binary classification
- Finding the optimal separation other than linear is a NP-Hard problem



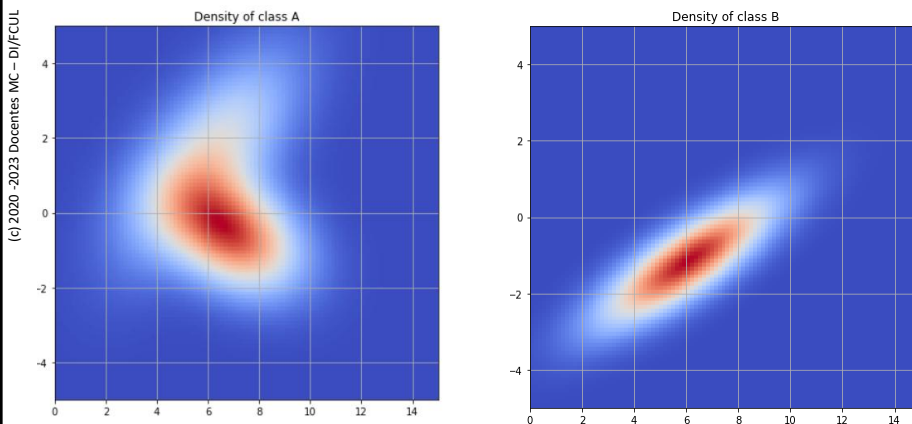
The idea of classification

- The optimal Bayes classifier can only be produced in artificial conditions where we know exactly the intrinsic characteristics of the populations to be classified
 - In this case these populations were generated as multivariate normal distributions (population A actually bimodal)
 - It is possible to define analytically the separating line between both distributions
 - This is the **Optimal Bayesian classifier**

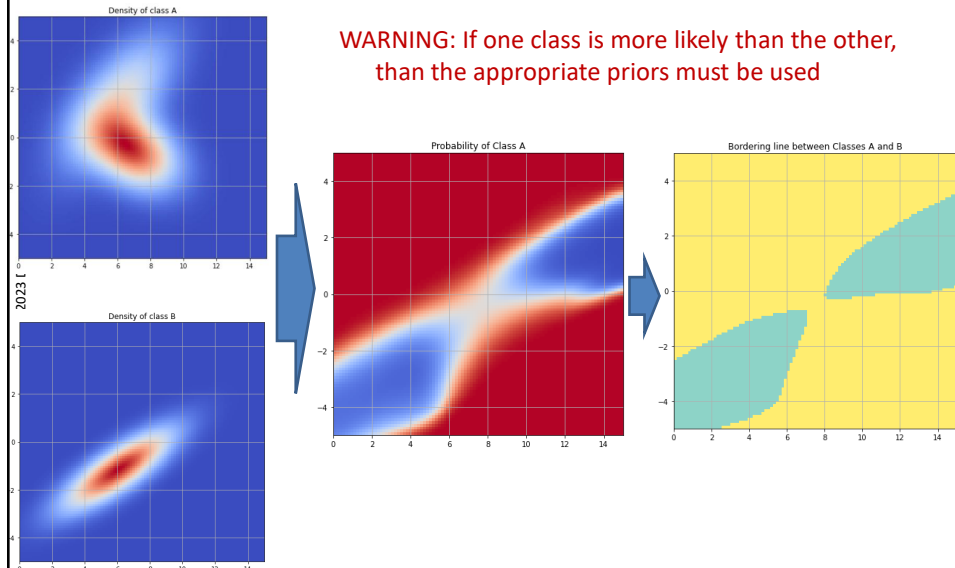


The idea of classification

- 2 classes: Class A and Class B



The Optimal Classifier



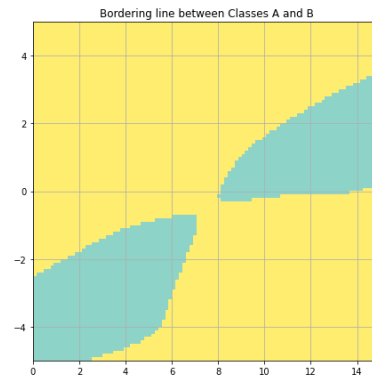
Optimal Bayesian Classifier

- An Optimal Bayesian Classifier (OBC) is NOT a perfect classifier
- **It is a model that produces the minimal theoretical error**
- The goal of any classification model is to approach the OBC
 - The misclassification errors should approach as best as possible the ones from the OBC
 - If they are larger, the model has a lot of bias and there is room for improvement in the model
 - If they are smaller, the model either has a lot of variance or suffers from any type of sampling issue => **it does not mean the model is "better"**
 - On any case, even if the errors of a model approach the ones of the OBC we may never truly approach the real dividing line

We must make clear that the OBC is a construct only possible in artificial data. In every real world problem we will never know the actual distribution of our data

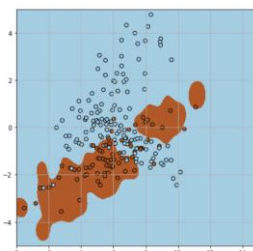
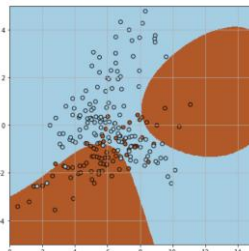
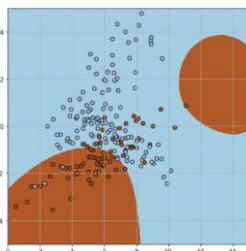
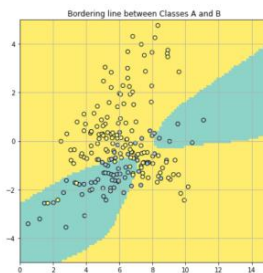
Universal Function Aproximators

- Theoretically if the actual distribution functions are known (and empirically if we have enough data) we should be able to find the perfect separation between classes
- We need a modelling tool capable of approximating any function delineation
 - **Universal Function Approximators**
- Many, but not all, Supervised Learning Tools are Universal Function Approximators
 - Able to learn from data the parameters that characterize each function
 - The shape of the basic curves and the fitting procedures are what differentiates between modelling families



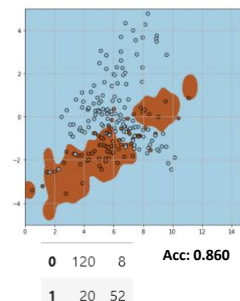
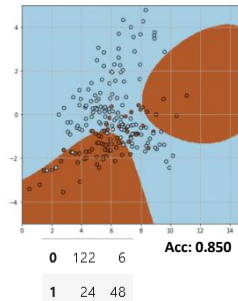
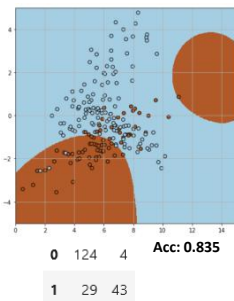
Finding the best model?

- Learning data
500 Points
- Testing data
200 points (plotted)
- Modelling tool:
 - A Support Vector Classifier



Results

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What does it all mean?

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- Is there a best model? (tricky question)
- Can we find the best model?
- Are we assessing each model accurately?
 - How confident are we of everything?
- Is each prediction equally good?
- Concrete questions
 - What if the training data was different?
 - What if the testing data was different?
 - What if the sample sizes were different?
 - What if we used [\[insert favourite modelling approach\]](#)?
 - Are we assessing the models correctly?
 - E.g. Are we taking care of the priors? [Class A is 2x more likely than class B]

A “real world” case

- [not really “real world”] –this is a well behaved problem and very easy
 - The Wine dataset 178 points and 3 classes [59, 71, 48]

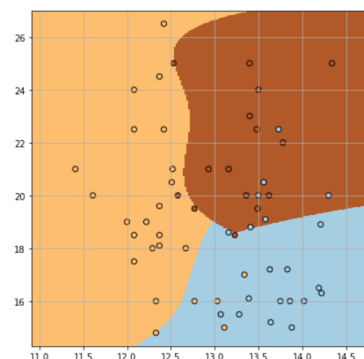
	0	1	2	3	4	5	6	7	8	9	10	11	12
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	0.28	2.29	5.64	1.04	3.92	1065.0
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050.0
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185.0
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480.0
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
173	13.71	5.65	2.45	20.5	95.0	1.68	0.61	0.52	1.06	7.70	0.64	1.74	740.0
174	13.40	3.91	2.48	23.0	102.0	1.80	0.75	0.43	1.41	7.30	0.70	1.56	750.0

- 1) Alcohol
- 2) Malic acid
- 3) Ash
- 4) Alkalinity of ash
- 5) Magnesium
- 6) Total phenols
- 7) Flavonoids
- 8) Nonflavanoid phenols
- 9) Proanthocyanins
- 10) Color intensity
- 11) Hue
- 12) OD280/OD315 of diluted wines
- 13) Proline

<https://archive.ics.uci.edu/ml/machine-learning-databases/wine/>

A 2D model

- A 2D model [Alcohol and Ash]
 - Training set: 119 instances
 - Testing set: 59 instances
- Having a 2D model is convenient but we have 13 variables.
 - Are these 2 (Alcohol and Ash) the best variables?
 - Can we use all of them in a model?
 - Should we?



	0	1	2
0	14	0	6
1	5	19	0
2	1	3	11

Acc: 0.746

A full model

- Same data
 - Same partition
- Same modelling approach
 - Default parameters
- Using 13 dimensions
- What is happening?
 - What can we do about it?

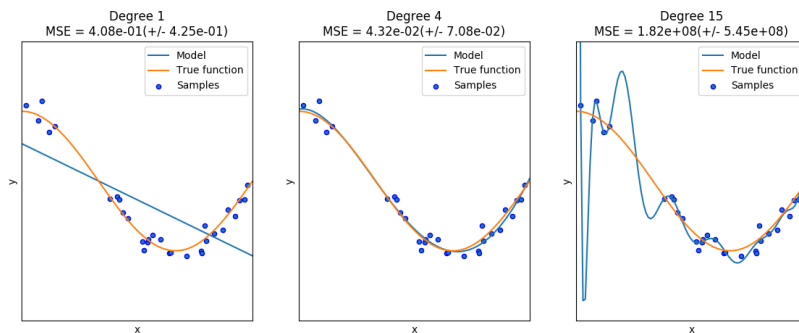
	0	1	2
0	20	0	0
1	0	15	9
2	0	8	7

Acc: 0.712

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The goal of regression

- The goal of any regressor is to make an approximation as best as possible to the actual function that generated the data
 - Again the need for a Universal Function Approximator
 - The shape of the curve may be able to extrapolate to other regions and domains



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A “real world” case

- The diabetes dataset
 - Originally 20 variables **transformed into 10** on scikit
 - 442 samples
 - Y ranging from [24 to 346]

	0	1	2	3	4	5	6	7	8	9
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019908	-0.017646
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068330	-0.092204
2	0.085299	0.050680	0.044451	-0.005671	-0.045599	-0.034194	-0.032356	-0.002592	0.002864	-0.025930
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022692	-0.009362
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031991	-0.046641
...	...	...	...	...	...	...	...	...	...	...
437	0.041708	0.050680	0.019662	0.059744	-0.005697	-0.002566	-0.028674	-0.002592	0.031193	0.007207

<https://archive.ics.uci.edu/ml/datasets/diabetes>

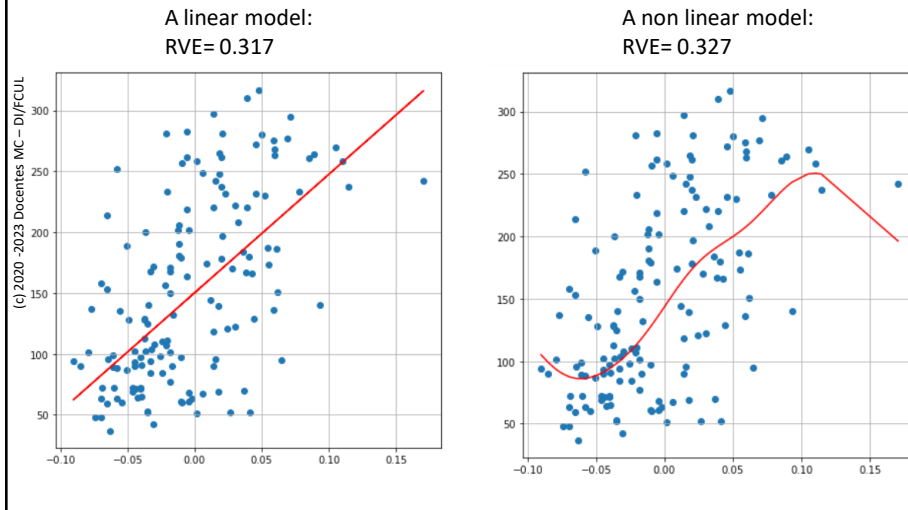
A 1 D model

- Choose the best independent variable
- One possibility: compute the R2 of each individual pair (x,y)
- Must assume a linear correlation! [does it make sense?]
- Steps:
 - Create a sample for testing and training
 - Compute the coefficient of determination (or proportion of the variance explained) **with the training data**
- Select the highest value
- “Best variable” is **variable 2**

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} = \frac{SS_{\text{reg}}/n}{SS_{\text{tot}}/n}$$

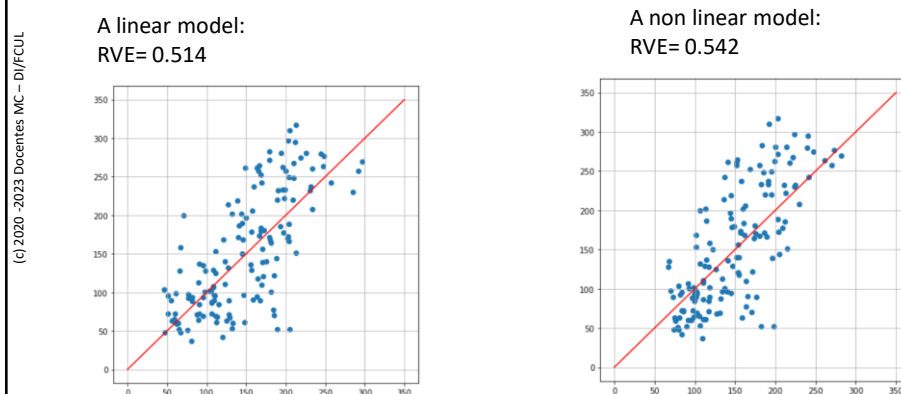
Var: 0	-	0.0304
Var: 1	-	0.0003
Var: 2	-	0.3557
Var: 3	-	0.2009
Var: 4	-	0.0291
Var: 5	-	0.0198
Var: 6	-	0.1506
Var: 7	-	0.1642
Var: 8	-	0.2595
Var: 9	-	0.1191

Choosing between models



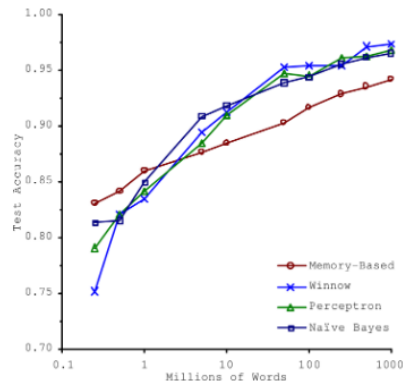
And using all variables

- Prediction vs Truth plots



The unreasonable effectiveness of data

- Banko and Brill in 2001 showed that different machine learning algorithms including the simplest ones performed very similarly in natural language problems if enough data was given
- Are all the problems we are facing consequences of not having enough data?
- Peter Norvig (2009) further popularized the topic in a famous paper but mentioned that many times big datasets are simply not available and, for most problems, we really must struggle with the data



<https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/35179.pdf>

To conclude

- Supervised classification is typically a difficult problem
 - Choosing the right algorithm is important but:
 - Processing the data is critical
 - Understanding the data is critical
 - Adequately evaluating the models is critical
 - And sometimes it is fundamental to have a realistic expectation of what can actually be accomplished

Elements of study

- James et al, ISLP, 2023 – Chapters 1 and 2
- Hastie et al, 2009 – Chapter 2
- Géron, 2019 – chapters 1 and 2

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Next class

- A linear data partition algorithm for classification and regression
 - Tree models!

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